

Lecture 8. Human Performance Modeling - 6

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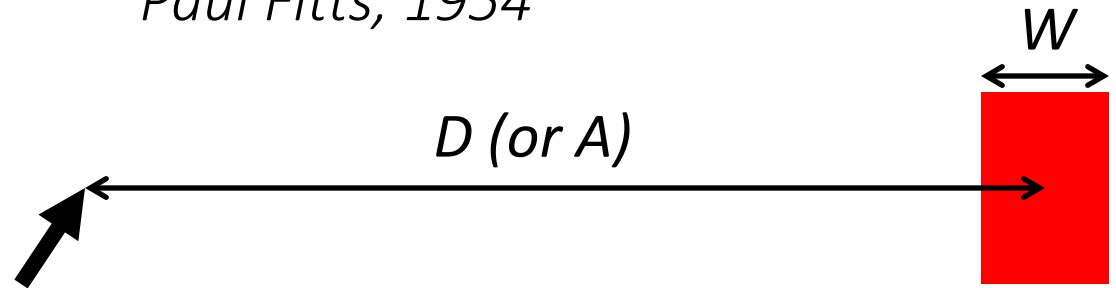
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Fitts' Law

Paul Fitts, 1954



$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

Movement Time

Index of Difficulty ($ID [bits]$)

The original Fitts' law proposed by Paul Fitts in 1954:

$$MT = a + b \log_2 \left(\frac{2D}{W} \right)$$

Crossing Law

- Not just pointing, how about crossing a boundary?



Johnny Accot and Shumin Zhai. 2002. More than dotting the i's --- foundations for crossing-based interfaces. CHI '02. ACM, New York, NY, USA, 73-80.

More than dotting the i's – Foundations for crossing-based interfaces

- Experiment result

$$CP: \quad T = 103 + 172 \times ID \quad r^2 = 0.998 \quad (4)$$

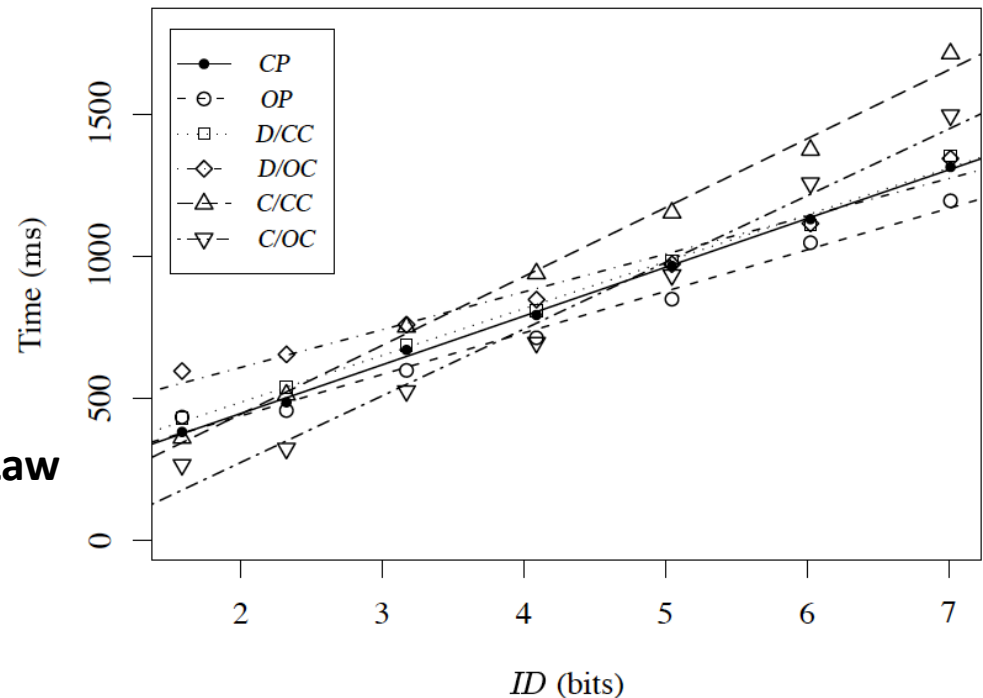
$$OP: \quad T = 145 + 146 \times ID \quad r^2 = 0.986 \quad (5)$$

$$D/CC: \quad T = 155 + 165 \times ID \quad r^2 = 0.994 \quad (6)$$

$$D/OC: \quad T = 342 + 133 \times ID \quad r^2 = 0.975 \quad (7)$$

$$C/CC: \quad T = -41 + 242 \times ID \quad r^2 = 0.995 \quad (8)$$

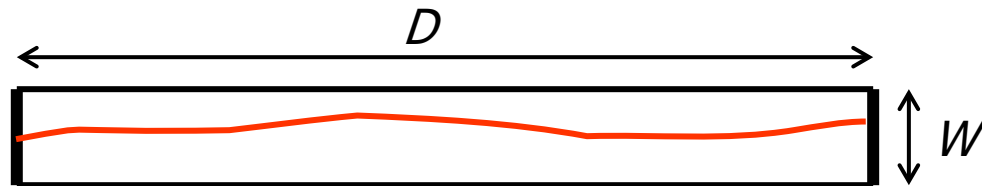
$$C/OC: \quad T = -196 + 235 \times ID \quad r^2 = 0.984 \quad (9)$$



Conclusion:
Crossing tasks can be modeled by Fitts' Law

Steering Law (a.k.a Accot-Zhai Law)

Johnny Accot and Shumin Zhai. 1997. Beyond Fitts' law: models for trajectory-based HCI tasks. In *CHI '97*. ACM, New York, NY, USA, 295-302.



$$MT = a + b \frac{D}{W}$$

$$ID = \frac{D}{W}, \quad MT = a + bID,$$

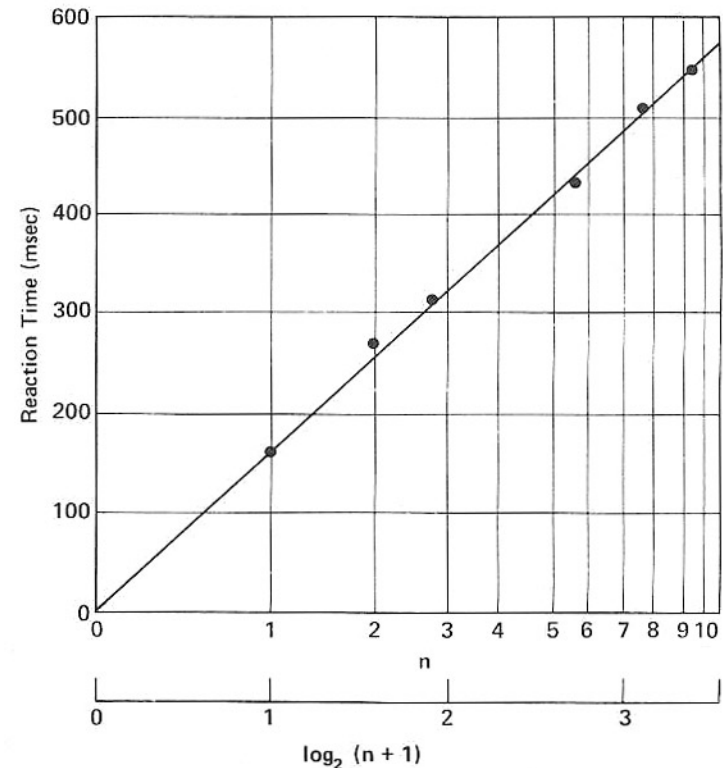
Choice Reaction Time

Hick's Law

The time it takes for a person to make a decision as a result of the possible choices he or she has: increasing the number of choices will increase the decision time logarithmically.

$$T = b \cdot \log_2(n + 1)$$

At the onset of one of n lights, arranged in a row, the subject is to press the key located Below the light (After Welford, 1968, p62)



Applications of Hick's Law in interface design

Contextual Menus: Right-click menus often show only the most relevant commands based on context, rather than the full set of options, to minimize decision time.

One-button design of iPhone



vs.

Three-button design of Android



Power Law of Practice

- Task time on the nth trial follows a power law

$$T_n = T_1 n^{-a} + c$$

where $a = .4$ [$.2 \sim .6$], c = limiting constant

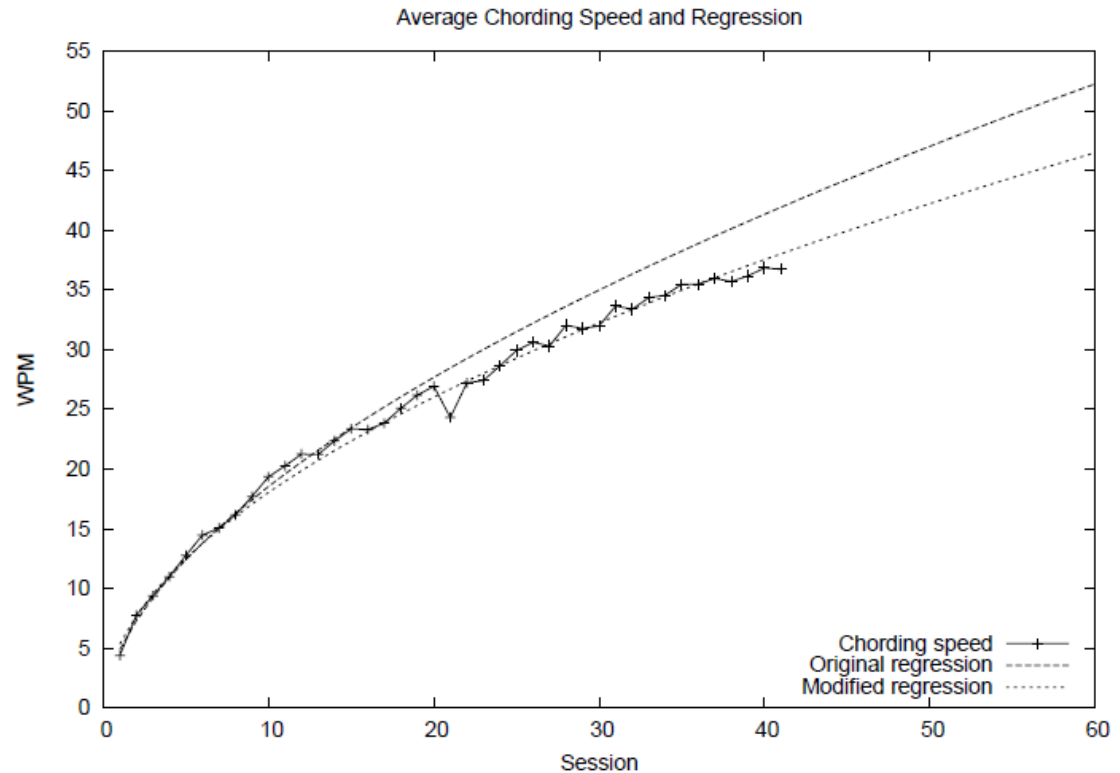
- You get faster the more times you do it!

Applies to skilled behavior (sensory & motor)

Does not apply to

- Knowledge acquisition

An Example of Power Law of Practice



Original regression : $y = 4.8987x^{0.5781}$, $R^2 = 0.9849$

Modified regression : $y = 5.3503x^{0.5280}$, $R^2 = 0.9787$

K. Lyons, D. Plaisted and T. Starner, "Expert chording text entry on the Twiddler one-handed keyboard," *Eighth International Symposium on Wearable Computers*, 2004, pp. 94-101

Keyboard Layout Learning

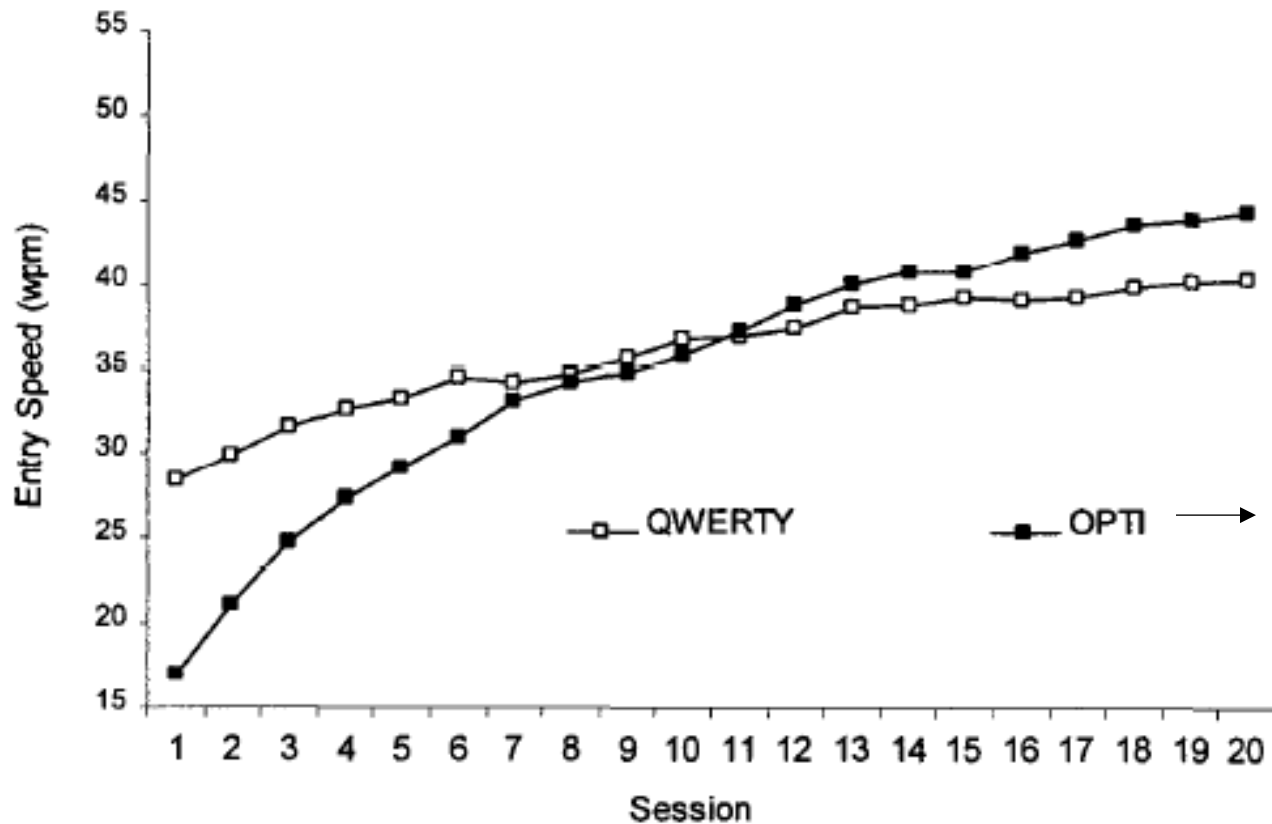


Figure 8. Entry speed by keyboard layout and session

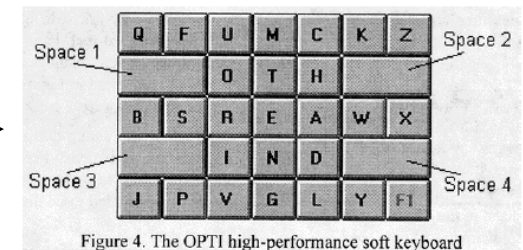
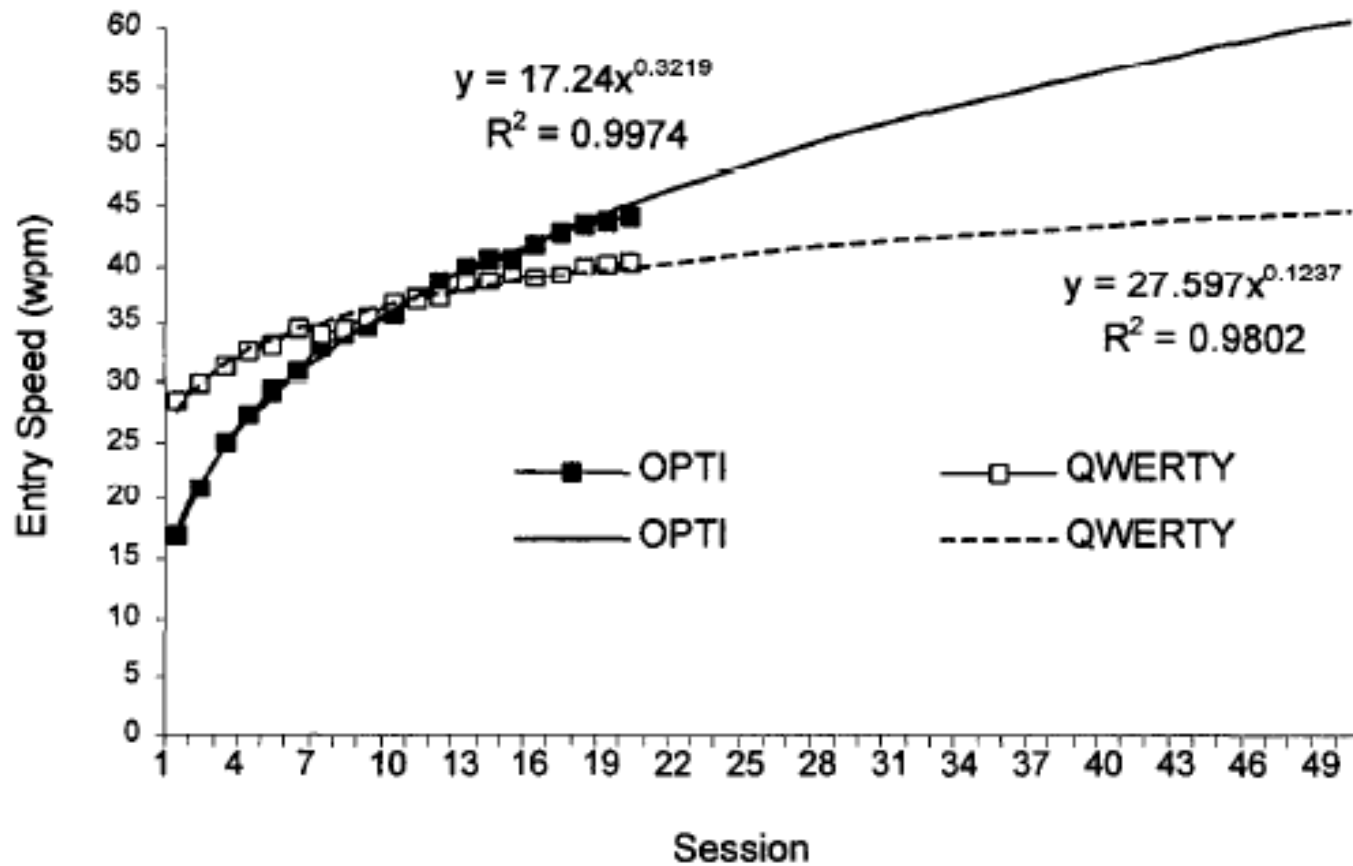


Figure 4. The OPTI high-performance soft keyboard

Keyboard Layout Learning

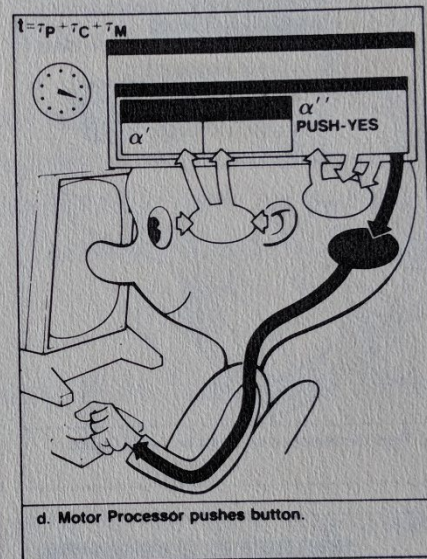
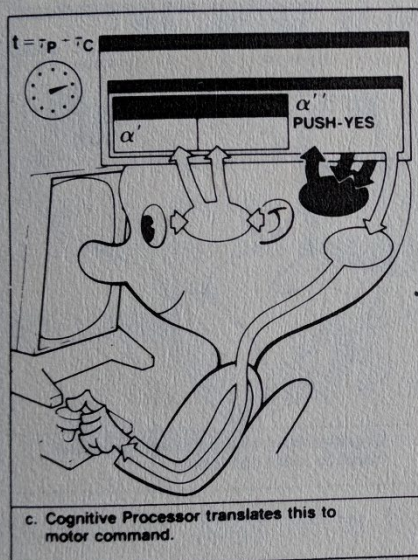
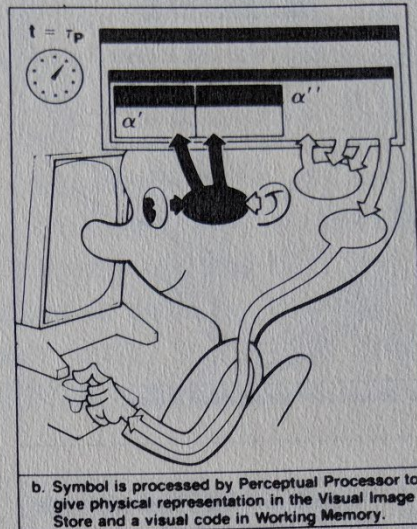
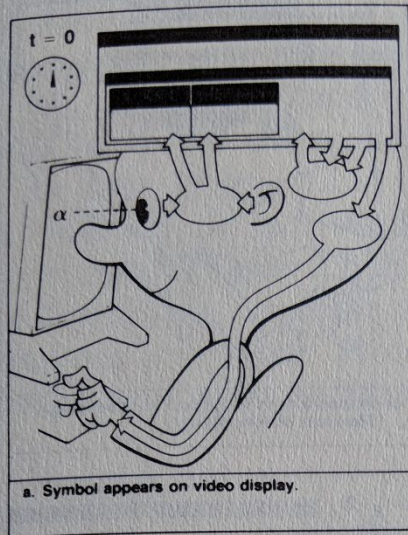


GOMS

Question?

- How much time will it take to replace all instances of a 4-letter word in MS Word?

A user sits before a computer display terminal. Whenever any symbol appears, he is to press the space bar. What is the time between signal and response?



Reaction Time =

$$\tau_p + \tau_c + \tau_M =$$

$$100 + 70 + 70 = 240 \text{ ms}$$

What is GOMS?

- A family of user interface modeling techniques
- Goals, Operators, Methods, and Selection rules
 - Input: detailed description of UI and task(s)
 - Output: various qualitative and quantitative measures

- Goals: These are the objectives that the user wants to achieve.
- Operators: These are the basic actions that the user performs.
- Methods: These are sequences of operators that accomplish sub-goals.
- Selection rules: These are the decision points that determine which method to use.

What can GOMS model?

- Task must be goal-directed
 - Not problem solving or creative tasks (e.g. design)
- Can include serial and parallel tasks

How to Do GOMS Analysis

- Generate task description
 - Pick high-level user Goal
 - Write Method for accomplishing Goal - may invoke subgoals
 - Write Methods for subgoals
 - This is recursive
 - Stops when Operators are reached
- Evaluate description of task
- Apply results to UI

GOMS Output

- Execution Time
- Functionality coverage and consistency
 - Does UI contain needed functions?
- Operator sequence
 - In what order are individual operations done?
 - Abstraction of operations may vary among models

An Example: Keystroke-Level Model

- How to make a KLM
 - List specific actions user does to perform task
 - Keystrokes and button presses
 - Mouse movements
 - Hand movements between keyboard & mouse
 - System response time (if it makes user wait)
 - Assign execution times to steps
 - Add up execution times
- Only provides execution time and operator sequence

KLM Example

- Replace all instances of a 4-letter word.

(example from Hochstein)

Description	Operation	Time (sec)
Reach for mouse	H[mouse]	0.40
Move pointer to "Replace" button	P[menu item]	1.10
Click on "Replace" command	K[mouse]	0.20
Home on keyboard	H[keyboard]	0.40
Specify word to be replaced	M4K[word]	2.15
Reach for mouse	H[mouse]	0.40
Point to correct field	P[field]	1.10
Click on field	K[mouse]	0.20
Home on keyboard	H[keyboard]	0.40
Type new word	M4K[word]	2.15
Reach for mouse	H[mouse]	0.40
Move pointer on Replace-all	P[replace-all]	1.10
Click on field	K[mouse]	0.20
Total		10.2

According to this KLM model, it takes 10.2 seconds to accomplish this task.

Example #2

Scenario:

A user wants to **copy and paste text** from one document to another document

Goal: copy a section of text and paste it into another document.

Operators

- M** = Mentally decide the method (~1.2s)
- P** = Pointing to an element using a mouse (~1.1s)
- B** = Clicking a mouse button (~0.1s)
- K** = Pressing a key on the keyboard (~0.2s)
- H** = Moving a hand between keyboard and mouse (~0.4s)

Total estimated time: 7.8s

Method 1: Mouse-Based (Right-Click Menu)

- 1.**M** - Mentally decide to copy text (~1.2s)
- 2.**P** - Move mouse to select text (~1.1s)
- 3.**B** - Click and drag to highlight text (~1.5s)
- 4.**B** - Right-click to open menu (~0.1s)
- 5.**P** - Move mouse to "Copy" option (~1.0s)
- 6.**B** - Click "Copy" (~0.1s)
- 7.**P** - Move mouse to second document (~1.5s)
- 8.**B** - Click in the document (~0.1s)
- 9.**B** - Right-click to open menu (~0.1s)
- 10.**P** - Move mouse to "Paste" option (~1.0s)
- 11.**B** - Click "Paste" (~0.1s)

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- H** = Moving a hand between keyboard and mouse (~0.4s)

Total estimated time: 7s

Method 2: Keyboard Shortcuts

- 1.**M** - Mentally decide to copy text (~1.2s)
- 2.**P** - Move mouse to select text (~1.1s)
- 3.**B** - Click and drag to highlight text (~1.5s)
- 4.**H** - Move hand to keyboard (~0.4s)
- 5.**K** - Press Ctrl + C to copy (~0.2s)
- 6.**H** - Move hand to mouse (~0.4s)
- 7.**P** - Move mouse to second document (~1.5s)
- 8.**B** - Click in the document (~0.1s)
- 9.**H** - Move hand back to keyboard (~0.4s)
- 10.**K** - Press Ctrl + V to paste (~0.2s)

- M** = Mentally decide the method (~1.2s)
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Advantages of GOMS

- Gives several qualitative and quantitative measures
- Model explains **why** the results are what they are
- Less work than usability study
- Easy to modify when interface is revised
- Research ongoing for tools to aid modeling process

Disadvantages of GOMS

- Only works for goal-directed tasks
- Assumes tasks are performed by expert users
- Does not address several important UI issues, such as
 - readability of text
 - memorability of icons, commands

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